Office of the President
Ground Floor, Research Tower Building
#1 Bolton Street Barangay 3-A, Poblacion, Davao City
Telefax No. (082) 221-7035
Phone No. (082) 227-5456 Local 105/110

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TREATISE ON THE USE OF ARTIFICIAL INTELLIGENCE AND OTHER ONLINE PROGRAMS IN THE UNIVERSITY OF MINDANAO[1]

As of June 10, 2024

Policy Statement

The University of Mindanao shall vigorously proceed and adopt education technology and innovations, including artificial intelligence and online programs, that are beneficial to the achievement of learning outcomes and performance excellence, but without sacrificing academic integrity. The ethical use of technology, tempered by available resources, will be a sacrosanct obligation of all sectors in the university.

Rationale

> *UNESCO has raised the alarm over using artificial intelligence tools in schools, urging regulators to implement tighter restrictions on their usage….The UN agency points to a spike in plagiarism and cheating incidents via AI platforms.*
>
> — Wahid Pessarlay in *The Business*, 12 September 2023

As opposed to human intelligence, artificial intelligence (AI) is generally referred to as machine intelligence based on a set of programmed and programmable tasks using various algorithms and developed in other online programs as software or computer systems, including search engines, chatbots, automated applications, or pattern recognition programs (facial, speech, movement, voice, language), among others. It is a technology designed to make computers or machines intelligent that are able to search and respond to queries or tasks on demand with speed; calculate or analyze problems; even learn and make decisions with abilities similar to or even better than humans. AI is intended for more efficiency in doing human tasks, so that humans can do other tasks that are difficult for computers to perform. For the use of commerce and industry, AI is seen as a technological panacea as part of industrial revolution, as it can learn and develop work processes. Other online programs with interactive features even with limited AI are also intended to help humans in their day-to-day tasks.

While the use of AI and other online programs can enhance organizational operational productivity, they can bring huge challenges to many pedagogical principles, policies, and practices. Without proper caution, orientation, mitigation, and control, these challenges

[1] Collaborative academic policy paper by Dr. Ronnie V. Amorado and Dr. Pedrito M. Castillo. This treatise is a living document and thus needs to be updated as AI and AI in education are not yet fully understood, explored, and utilized as a tool.

can bring about distortions that hamper, mock, or even injure educational standards and requirements expected of all learners and learning facilitators (academic and academic support personnel). No less than the UNESCO[2] and several scholars[3] have raised the red flags against the abuse and use of AI in education and thus, require schools to design ethical framework and internal policy guidance for all its stakeholders.

This treatise is generally aimed to provide a set of simple and practical guidelines on the use of AI and other online programs by all the stakeholders in the university. It is the desire of the university to take stock of the opportunities offered by these technological applications but at the same time be able to prepare to face the challenges they pose to the university. It is important to highlight that UM is an educational institution that seeks to mold and cultivate student learning competencies and outcomes. Thus, this treatise shall continue to uphold the basic and traditional learning principles while adapting to new technologies.

Situationer[4]

> *Dishonest students will cheat more in online classes. But even the honest students are also tempted – even seduced – because of the opportunity and ease of cheating. The nature of the online technology presents daunting problems for systems integrity (pedagogy) and the moral integrity of the users (teachers and students).*
>
> — Craig Markovitz[5]

Academic fraud or more specifically — cheating in online classes — have become unrestrained and unchecked during the online era. This is made possible due to the fact that students are not prepared and many lack self-governance and self-discipline for self-directed learning during their online classes. The teachers are also incapable of gauging, checking, and even preventing the occurrence of cheating, especially in online examinations. Craig Markovitz wrote in *The Degree 360,* "*online courses and exams certainly make it easier for dishonest students to cheat*."

What Craig Markovitz wrote was part of several commentaries about cheating in online classes in the United States. As online education is already more than a $100-billion global industry in 2015 and projected to peak at $350-billion by 2025 (pre-pandemic estimates), cheating in online classes is now poised to be a big business for the enterprising. Some following empirical data in the US provide the proof.

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[2] United Nations Educational, Scientific and Cultural Organization (UNESCO). 2022. "Recommendation on the Ethics of Artificial Intelligence." UNESCO: Paris, France.

[3] Most recently, Dr. Christian Ryan Maboloc published Chat GPT: the need for an ethical framework to regulate its use in education, *Journal Of Public Health*, Volume 46, Issue 1, March 2024, Page 152 @ https://doi.org/10.1093/pubmed/fdad125.

[4] The situationer was culled from the latest research and book publication by Ronnie V. Amorado & Ramcis N. Vilchez. 2024.*Pedagogy of Fraud: Cheating Practices in Online Classes*. ISBN 978-971-98-2304-9. Philippines: C & E Publishing, Inc.

[5] Markovitz, Craig. The Degree 360, April 26, 2019.

A higher education survey in the US with more than 2,000 college professors and administrators revealed that about 60% believed that academic fraud is more common in online courses than in face-to-face courses.[6] The survey results were similar to those reported by King, et al. (2009)[7] wherein 73.8% felt it was easier to cheat in an online class versus a traditional one. Moreover, the study was indicative of tolerant views of probable cheating behaviors when the teacher does not set a test-taking policy. In a study about online schools in the US from a survey of more than 600 undergraduate and graduate students, about 33% admitted to cheating in online classes but only 2% were caught cheating. This is the reality of the nature of online technology; it is vulnerable to cheating and it is difficult to get caught.

Over the last decade, online learning has become a prominent industry. It has shown substantial growth as the Internet and education converged to provide a platform for people to learn new skills. Even before the pandemic, Research and Markets forecasted that the global online education market would reach $350 Billion by 2025 (Koksal, 2020)[8]. With the increase in online college education, cheating is a booming business, where students have more opportunities to outsource their schoolwork (Newton, 2020).[9]

Similarly, among 639 undergraduate and graduate students from two universities, results show that cheating happened more frequently in online classes for students enrolled in both online and face-to-face classes (Miller and Young-Jones, 2021).[10] Another study discovered that among the respondents of 635 undergraduate and graduate students, 32.7% admitted to cheating in online classes, and 32.1% admitted to cheating in live classrooms. In comparison, only 2.1% of online students and 4.9% of students in a live class were caught cheating (Watson and Sottile, 2010).[11] Data suggests that students were almost four times more likely to cheat in online classes than in live classes and that their classmates were over five times more likely to cheat (Watson and Sottile, 2010). In addition, Yardley et al. (2009)[12] earlier surveyed the prevalence and perceived severity of cheating behaviors of several alumni from different universities and found that most of the respondents (81.7%) have reported that they committed at least one form of cheating behavior during their undergraduate years. Also, the study affirmed that "*copying from another student's assignment*" and "*allowing others to copy from your assignment*" were the most common forms of cheating, and students' top reasons for cheating were lack of time and to help a friend.

[6] Newton, Derek. Forbes Magazine, October 30, 2019.
[7] King, C.G, et al. (2009). "Online exams and cheating: An empirical analysis of business students' views." *Journal of Educators Online*, 6(1), n1.
[8] Koksal, I. "The Rise Of Online Learning." May 2, 2020
[9] Newton, Derek. Another problem with shifting education online: cheating. *Hechinger Report.* August 7, 2020.
[10] Miller, A., and A.D. Young-Jones. (2012). "Academic integrity: Online classes compared to face-to-face classes." *Journal of Instructional Psychology*, 39(3).
[11] Watson, G. and J. Sottile. (2010). "Cheating in the Digital Age: Do Students Cheat More in Online Courses?" *Online Journal of Distance Learning Administration* 13.1 (2010).
[12] Yardley, J., M.D. Rodríguez, S.C. Bates, and J. Nelson. (2009). "True Confessions? Alumni's Retrospective Reports on Undergraduate Cheating Behaviors." *Ethics & Behavior*, 19(1), 1–14. doi:10.1080/10508420802487096

Over the years, students get creative and learn many new ways to cheat with the help of Internet technologies. Rowe (2004)[13] cited three common ways students used dishonesty in online assessment: taking their exams later to gather leaked answers; retaking exams based on false assertions; and getting unauthorized help during exams. A Kessler student survey[14] in the US in 2017 revealed the following:

1. 76% said they copied texts from somebody else's assignment;
2. 79% admitted to plagiarism from Internet sources;
3. 72% said they used mobile devices to cheat;
4. 42% said they purchased custom papers or essays online; and
5. 28% said they paid for a service to take their online classes for them.

In a MacAfee study, 1 in 3 American students use mobile phones or other connected devices to cheat on exams.[15] This number shows that 30% of American students are cheating in their online classes. This figure affirmed the survey of online schools showing about 33% of students admitting to cheating in online class. How do American students cheat? Some trends of cheating practices were identified:

1. Online identity fraud (identity spoofing; somebody is attending the online classes);
2. Impersonation (small-scale identity fraud);
3. Surrogacy (large-scale impersonation);
4. Outsourced assignments;
5. Plagiarism;
6. Call a friend (during exams); and
7. Consult Google (during exams)

In her own paper, Hollis (2018)[16] found that students are increasingly becoming savvy in cheating in their online classes, including buying so-called allies to serve as ghost-students to take an entire class or examination for them. This is similar to impersonation and surrogacy. Citing other authors, Hollis acknowledged that an increasing number of students felt that it was easy to cheat online. The figures grew from 63% in 2009 to 74% in 2017. Hollis likewise noted the opportunity to cheat in the online environment, driven most especially by anonymity.

Hughes and McCabe (2006)[17] suggest possible indicators of academic misconduct: student maturity; perceptions of what constitutes academic misconduct; faculty assessment and invigilation practices; low perceived risk; ineffective and poorly understood policies and procedures; and a lack of education on academic misconduct.

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[13] Rowe, N. C. (2004). "Cheating in online student assessment: Beyond plagiarism." *Online Journal of Distance Learning Administration*, 7(2).

[14] Kessler International: Survey shows cheating and academic dishonesty prevalent in colleges and universities. 2017.

[15] Romila Kanchan, Online Remote Proctoring, May 11, 2020.

[16] Hollis, Leah. (2018). "Ghost-Students and the New Wave of Online Cheating for Community College Students." *New Directions for Community Colleges* No. 183. USA: Wiley Periodicals. 2018(183), 25-34.

[17] Hughes, J. M. C. and D.L. McCabe. (2006). "Academic misconduct within higher education in Canada." *Canadian Journal of Higher Education*, 36(2), 1-21.

At a Congressional hearing, former Inspector General Kathleen S. Tighe of the US Department of Education testified on why American students are cheating:[18]

> [The] "*management of distance education program presents a challenge…. because of limited or no physical contact to verify the students' identity or attendance. [This is] because of aspects of distance education that take place through the internet; students are not required to present themselves in person at any point*."

Tighe's testimony is very crucial in understanding better the nature of online technology. Both Derek Newton and Leah Hollis were correct in noting that anonymity and distance present as drivers for students to cheat. Dishonest students, as Craig Markovitz would describe, will cheat more in online classes. But even the honest students are also tempted — even seduced—because of the opportunity and ease of cheating. The nature of the online technology presents daunting problems for its systems integrity (pedagogy) and the moral integrity of the users (teachers and students).

In the other parts of the world, cheating in online classes is pervasive. Ayoub/Al-Salim and Aladwan (2021)[19] found cheating by students in online classes are increasing in recent years as technologies are abused in the following areas – cutting and pasting materials from the Internet; sharing online quizzes; and texting answers to classmates. Online technologies have also become the platform for the conception of a new set of cheating practices called e-cheating or e-dishonesty[20] (Holden, et al., 2021).[21] Citing various sources, Holden, et al. (2021) explained the forms of e-cheating and academic dishonesty as including the following examples:

- downloading papers from the Internet and claiming them as one's own work;
- using materials without permission during an online exam;
- communicating with other students through the internet to obtain answers;
- having another person complete an online exam or assignment rather than the student who is submitting to work;
- using unauthorized materials or forbidden resources during the exam;
- facilitation (helping others to cheat);
- falsification (misrepresentation of oneself);
- plagiarism (claiming another's work as one's own); and
- providing an unearned advantage over other students

[18] Kathleen S. Tighe, US Department of Education Congressional Testimony, March 19, 2013.

[19] Ayoub/Al-Salim, Majda and Khaled Aladwan. (2021). "The relationship between academic integrity of online university students and its effects on academic performance and learning quality." *Journal of Ethics in Entrepreneurship and Technology*, Vol 1. Emerald Publishing Limited.

[20] Holden, et al. (2021:4) describes the growing e-cheating culture in online classes, to wit: "*By some accounts, a primary contributor to academic dishonesty is the existence of e-cheating culture. If a university has an established culture of cheating – or at least the perception of a culture of cheating – students may be tolerant of cheating, believe that cheating is necessary in order to succeed, and believe that all students are cheating. Students directly shape cheating culture, and thus subsets of students in a university population may have their own cheating cultures. It is possible, then, for online students to have their own cheating culture that differs from the rest of the student population*."

[21] Holden, Olivia et al, (2021). Academic Integrity in Online Assessment: A Research Review. ON, Canada: Department of Psychology, Queens University.

Holden, et al. (2021) likewise describes e-dishonesty as referring to “behaviors that depart from academic integrity in the online environment.” In explaining why students resort to e-cheating and become e-dishonest, Holden, et al. (2021) cited the so-called fraud triangle as propounded in various literature. The fraud triangle considers three major factors:

- opportunity (students perceive that they can cheat without being caught)
- incentive, pressure or need (coming from parents, peers, friends)
- rationalization or attitude (perceives that cheating is an acceptable norm)

Further, Holden, et al. (2021) did a comprehensive systematic review of various literature on academic integrity and cheating practices in online classes, and concluded that approximately 42-74% of students believe that it is easier to cheat in online classes. That is about four to seven students out of 10 perceiving the online technology as easy to cheat on. The range of estimation confirms the 60%-figure on the same data from the US surveys.

What have been done so far? Hollis (2018) cited some recommendations to control online cheating at the organizational and faculty level. The organizational solutions are the following: check student's IP addresses; use test centers for assessment; student report in person to the specific test center with identification on hand; offer hybrid courses; clearly define what academic dishonesty is and impose severe penalties for ghosting; and hire personnel that deals with possible ghosting, instead of faculty. Meanwhile, the faculty-level recommendations are the following: require photo ID for registration and during online exams; require students to submit course notes and drafts of work; provide links to the anti-plagiarism program; utilize webcams in supervising exams or quizzes; and require all professors to include academic integrity policies on their course syllabus.

It appears that having a code of honor is far less effective in delivering online courses and leads to only a negligible decrease in online cheating behaviors (Corrigan-Gibbs et al., 2015).[22] Thus, the authors propose incorporating stern warnings that inform the students of the potential consequences of cheating if committed. This method leads to a vital drop in cheating. Furthermore, Bilen and Matros (2021)[23] propose two implementable solutions for the online cheating problem: an online exam policy that requires capturing each student's computer screen and room, and creating exams with more straightforward questions with less time to answer. On the other hand, Rowe (2004) suggests additional ways to minimize e-cheating, such as using exam software to generate randomized questions, safeguard test bank accessibility, and produce various versions of each assessment. In support of the latter investigation, Golden and Kohlbeck (2020)[24] validated paraphrasing test bank questions to minimize cheating on online exams. Results show that students perform better on verbatim questions than paraphrased ones (80.4% vs. 69.1%).

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[22] Corrigan-Gibbs, H., N. Gupta, C. Northcutt, E. Cutrell, and W. Thies. (2015). “Deterring cheating in online environments.” *ACM Transactions on Computer-Human Interaction* (TOCHI), 22(6), 1-23.
[23] Bilen, E. and A. Matros. (2021). “Online cheating amid COVID-19.” *Journal of Economic Behavior & Organization*, 182, 196-211.
[24] Golden, J. and M. Kohlbeck. (2020). “Addressing cheating when using test bank questions in online classes.” *Journal of Accounting Education*, 52, 100671.

In the Philippine setting, a seminal research by Ronnie V. Amorado and Ramcis N. Vilchez reveal very disturbing trends.

An alarming 6 out of 10 in the regional survey and 5 out 10 in the nationwide survey admitted to cheating in their online classes. About the same number of students said that cheating in online classes is easier as compared to the residential, face-to-face classes. Further, about 8 out of 10 students admitted cheating in their online classes in various frequencies and up to 7 out 10 students are never caught cheating in their online classes. Majority are also aware of others who are cheating in online classes. One of the reasons why cheating can be pervasive in online classes is the un-detection factor, or the certainty of not being caught. Obviously, when the opportunity to cheat is reinforced by the lack of detection, the students tend to be audacious to cheat. Opportunity becomes more attractive and seductive without detection. The system of the online modality presents the attractive and seductive opportunity for students to cheat or be compelled to cheat. The favorite modes of cheating or *modus operandi* (MOs) for both the regional and nationwide surveys are:

1. Copying from Internet sources
2. Googling during quizzes and exams
3. Copying from somebody else's assignment
4. Calling a friend or classmate during quizzes and exams

These MOs resemble the patterns in the cheating practices in online classes in the United States.

Most easy to cheat in terms of assessment type for both surveys are multiple choice questions (MCQs), true or false, and matching types. Most difficult to cheat are essays and oral tests or examinations. In the online mode, objective types of assessment — especially the MCQs — are the most vulnerable to cheating. The demonstration type of assessment — those that belong to the so-called authentic assessment types — such as essays and recitations are the most difficult. However, there are software applications now that make essays even susceptible to fraud.

Cheating chat groups among the students are widely pervasive, with technology as the enabling factor that creates the seductive opportunity for students to cheat in online classes. The technology for online platforms are misused or abused, departing from their intended good use.

Further, the factors that emerged why students cheat are behavioral in nature, followed by assessment designs. These are the areas for possible intervention to strengthen students' values formation while improving the assessment designs that are appropriate for online delivery.

<u>Guiding Principles</u>

Even the wise cannot see all ends.
— J.R.R. Tolkien (1892-1973)

Responding to the current situationer and deriving from the university's core values of excellence, honesty and integrity, innovation, and teamwork as popularly expressed in 4G (*galing*, *gawi*, *gawa*, *gana*), this treatise articulates a set of guiding principles in the use, utilization, and exploitation of AI and other online programs for the perusal of all the university's stakeholders:

10. **Academic integrity** is an absolute guiding principle in using AI and other online programs in all the academic and non-academic operations in the university. This is intended to ensure that learners are molded and cultivated expectedly of their learning competencies. Technology is not intended for use to achieve delegated competencies. Learning facilitators – including the non-teaching personnel – are likewise expected to imbue such virtue as role models to learners.

11. **Technological adaptability** is derived from excellence and innovation that require the university to be continuously updated with latest technological trends and being adaptable to engage and adopt such technology to the extent that it is beneficial in achieving learning competencies (learners) and operational efficiency (learning facilitators).

As such, given the two guiding principles serving as foundational edicts, specific guiding principles are herein articulated in addition:[25]

12. AI and other online programs can be beneficial and thus are encouraged in terms of academic instruction, but must be dealt with extreme caution, if not proscription, in academic assessment.

13. Online lectures have limitations; explore online delivery strategies that will promote students' participation online such as online group workshops, online case discussions, or demonstrations. Online teaching is conducive for HOTS (higher-order thinking skills) such as analytical and problem-solving skills.

14. Strengthen the core values of integrity, self-governance, and self-discipline among the studentry. Online classes demand self-directed learning, and these core values serve as the effective pillars. Academic integrity is always paramount in the academic formation of students. Without integrity, academic excellence only promotes impunity. Without integrity, Filipino students can turn out to be the smartest and brightest cheaters and go scot-free. Imagine the effect when they become professionals, businessmen, government officials or employees, and worse — teachers in the future.

[25] Some of the principles adopt the recommendations from the study of Amorado and Vilchez (2024).

15. Do not treat the online classes – including those in Learning Management Systems (LMS) – as a fixed modality held in fixed space and time. There is a need to understand and harness synchronous and asynchronous learning. Many teachers treat online classes like regular face-to-face classes, thus monopolizing the online period with all the video conferencing lectures.

16. In the conduct of online classes, focus on giving feedback and facilitating interactive learning. Reduce the usual one-way lectures.

17. Performance assessment tasks (essays, problem solving, computational, demonstrations, etc) that are done face-to-face with teachers' monitoring are more reliable compared to online or traditional assessment tasks like MCQs or FCQs.

18. Use authentic assessment to motivate students and reduce student cheating in online classes and examinations. Students are also less likely to cheat when they are encouraged to demonstrate learning in ways that are most authentic to them; the key is motivation. Authentic assessments are useful in remote integrity and accountability.

19. Gauge student performance through an assessment spread in the entirety of students' online participation over time (use of varied question types). Authentic assessments for final examinations are ideal for online classes.

20. Discourage the use of MCQs/FCQs (forced choice questions), True-or-False, and Matching Types in online exams; they are the easiest to cheat. In fact, it is better to use more performance tasks than MCQs or FCQs especially in summative assessments.

21. Instructional assessment needs to become stronger and stringent to ensure originality and detect plagiarism using AI. Plagiarism is always a form of academic malpractice – including AI plagiarism – and must be dealt with accordingly.

22. AI in operations may also be useful especially in routinary and multi-tasking conditions, but professional intervention is needed for judgement. Operational outputs cannot just be solely left for AI alone.

23. Continuous learning and adaptation with AI through faculty and staff retooling is essential as part of pedagogical improvement.

AI in Education

Destroying any nation does not require the use of atomic bombs or the use of long range missiles. It only requires lowering the quality of education and allowing cheating in examinations by the students. Patients die at the hands of such doctors. Buildings collapse at the hands of such engineers. Money is lost at the hands of such economists and accountants. Humanity dies at the hands of such religious scholars. Justice is lost at the hands of such judges. The collapse of education is the collapse of the nation.

— Nelson Mandela (1918-2013)

The advent of AI, particularly the generative AI such as ChatGPT, Cici, Merlin, Bard, Nova, Copilot, Chatbots, Rufus, DeepDream and GitHub among others, has profoundly challenged the teachers' assessment of students' performance against expected learning outcomes. Generative AI definitely has decongested the academic and research tasks of students specifically in acquiring basic and advanced knowledge of any subject or topic of any discipline. Thus, there is a need to be mindful of AI capabilities in designing both knowledge and performance-based assessment tasks specifically in rationalizing the context (professional or real-world) of the task. Logically, generative AI implies that teachers should assign lesser weight to knowledge-based assessment tasks alone and considerable weight to performance-based or a combination of knowledge, skills and attitudes assessment tasks.

Reference Table for evaluating alignment of TLA, assessments and learning outcomes and integrating AI tool capabilities and distinctive human skills[26]

Bloom's Taxonomy Level	AI Capabilities	Distinctive Human Skills
Create	Suggest a range of alternatives, enumerate potential drawbacks and advantages, describe successful real-world cases	Formulate original solutions incorporating human judgment, collaborate spontaneously
Evaluate	Identify pros and cons of various courses of action, develop rubrics	Engage in metacognitive reflection, holistically appraise ethical consequences of alternative courses of action
Analyze	Compare and contrast data, infer trends and themes, compute, predict	Critically think and reason within the cognitive and affective domains, interpret and relate to authentic problems, decisions, and choices
Apply	Make use of a process, model, or method to illustrate how to solve a quantitative inquiry	Operate, implement, conduct, execute, experiment, and test in the real world; apply creativity and imagination to idea and solution development

[26] Artificial Intelligence Tools @ https://ecampus.oregonstate.edu/faculty/artificial-intelligence-tools/blooms-taxonomy.

Bloom's Taxonomy Level	AI Capabilities	Distinctive Human Skills
Understand	Describe a concept in different words, recognize a related example, translate	Contextualize answers within emotional, moral, or ethical considerations
Remember	Recall factual information, list possible answers, define a term, construct a basic chronology	Recall information in situations where technology is not readily accessible

Some strengths and weaknesses of AI in Education

Strengths	Weaknesses
Idea generation	Creative demonstrations of learning: posters, 3D projects, drawings, multimedia
Explain new or confusing concepts	Reflection activities
Precheck assignments for editing and feedback suggestions	Hands-on activities
Debate with an AI text generation tool asking it to take on a particular position	Applying learning to real-world scenarios
Text summaries	Collaborative learning
Refine AI prompts to improve the final output, for accuracy, persuasiveness, etc. rework and add to it	Open-ended discussions that require synthesis of course learning materials
Apply a concept to a new situation	Project-based learning
Ask for counter-arguments to a student-generated thesis or idea	Field trips followed by writing observations
Identify potential risks or causes of failure	Conduct interviews with industry experts

AI in Research

Plagiarism is the capital crime in academics.
— Dr. John M. Barrie, Turnitin founder

AI can be a useful research tool like other research tools used in statistics such as Excel, SAS, or SPSS, and data analytics like NVivo, Data Visualization, Python, KNime, Monte Carlo, RapidMiner, and Looker among others. As a technology tool, it can also be misused or abused. Thus, caution must be taken to adhere to academic integrity and technological adaptability without seducing malpractices in research. Some of the strategic directions can be pursued:

1. As part of technological adaptability, research may start developing AI skills in various research domains in the area of methodology and analysis.

2. As AI can be one of the powerful research tools, students and researchers need to be taught of the derivation to understand the process. This is similar to the use of statistical and data analytics tools.

3. Research needs to become stronger and stringent to ensure originality and detect plagiarism using AI in research outputs. The schools may adapt current originality thresholds using Turnit-In or other plagiarism checkers to include AI plagiarism.

4. AI cannot replace researchers, writers, or authors. AI programs and tools cannot be listed as researchers, writers, authors or co-authors.[27]

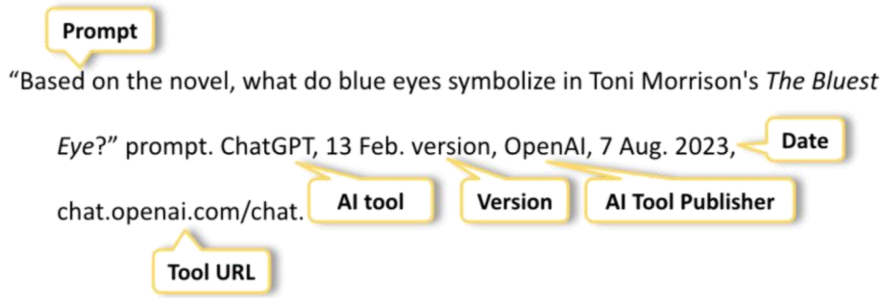
5. AI can be used as a reference or a source of information, similar to the techniques in writing review of related literature (RRL). Thus, a proper protocol must be adopted in citing AI sources.

[27] University of the Philippines – Los Baños Graduate School

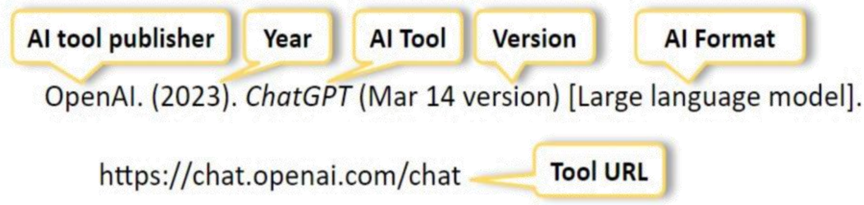
6. As an RRL, AI can be cited properly. For proper AI citation, the following must be included: prompt, AI tool, version, AI tool publisher, date accessed, and tool URL (not http)[28]. For example:

   OpenAI. (2023). ChatGPT (version 3.5). Conversation series from March 21–December 24, 2023 @ chat.openai.com.

   AI in MLA:



   AI in APA:



7. For in-text citation, AI can be cited as part of narrative citation: OpenAI (2024)…. or as parenthetical citation: ….. (OpenAI, 2024). However, due to reliability issues, it is strongly suggested that researchers, authors, teachers, and students shall be trained not to automatically cite AI. It is better to dig deeper and validate the responses or data from the AI, or if AI referred or cited authors, go to the authors themselves to find out. Classical journals or Google Scholars – which are also referenced or cited by AI – are still more reliable than just AI. For now, the tendency of AI is just like the Google Search or any search engine. It still very dependent on the database it is using. Research diligence is part of academic integrity.

8. When AI is used for referencing or citing other authors or sources, it is much better to investigate further, dig deeper, and look into them. Eventually, it is also more prudent to cite them directly rather than the AI responses. AI can still be hounded by reliability issues in terms of originality. No one knows if AI is telling the truth or not. Citing AI must be used sparingly.

9. There are some old-school, manual techniques to detect and prevent plagiarism, which are very effective up to the present juncture. Educators and researchers need to be trained on these techniques to help supplant online or AI plagiarism malpractices. In *Literati Dos*,[29] Ronnie Amorado propounded the following techniques to address plagiarism:

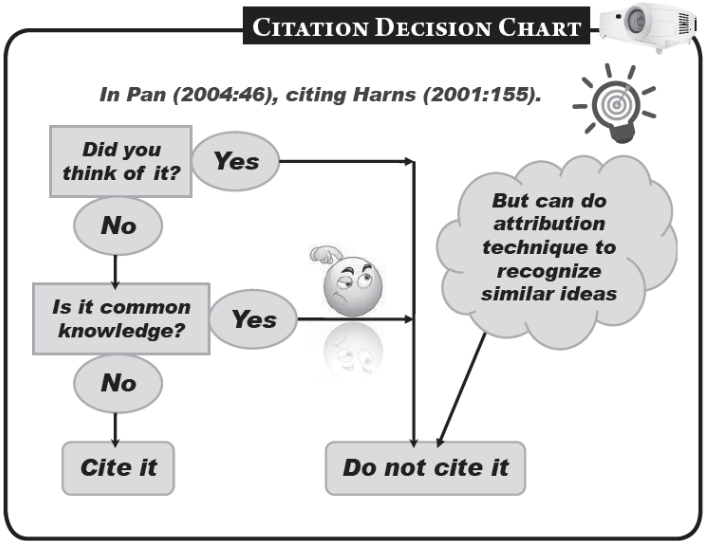
[28] University of Wisconsin Whitewater Libraries

[29] Amorado, Ronnie V. 2023. *Literati Dos. Literature Review in Contemporary Purview*. Manila: Mutya Publishing.

(a) Develop the habit of re-phrasing or re-stating sentences. It is a good practice to attribute the sources of literature (attribution technique). *Literati Dos* is advocating for the use of single and multiple collation techniques to prevent unintentional plagiarism.

(b) Develop the habit of using quotation marks for direct quotations or using exact sentences from another source. Make sure to cite (citation technique).

(c) Develop the habit of attributing and citing sources, even if they are re-phrased or re-stated already.



(d) Be guided by the Citation Decision Chart[30] on when to cite or not. Common knowledge usually does not require citation and thus cannot be plagiarized.

(e) Use the Rule of 10 by manual controls in detecting plagiarism: compare the cited authors in the main text with those entered in the references or bibliography at the back. If the uncited authors or sources reach 10, suspect of plagiarism.

(f) Check construction consistency. If some paragraphs are written in very good construction but others are glaringly badly written, suspect of plagiarism. Construction tenors serve as good manual detection.

(g) As part of manual detection, require the students or researchers to bring the books, journals or the materials they used as sources and check if properly cited.

(h) For intentional plagiarism, especially with significant volume of plagiarized entries, overhaul and re-write. Punish habitual, cunning or devious intentional plagiarists.

(i) Use a combination of manual detection (controls) and digital detection. There are several useful digital controls using antiplagiarism softwares such as Plagiarisma (Google Scholar), Plagiarism Checker (Grammarly), Turnitin, UniCheck, iThenticate, Viper, CopyGator, HelioBLAST, Tin Eye (for images or pictures) and many others.

[30] Pan, M. Ling. 2004. *Preparing Literature Reviews: Qualitative and Quantitative Approaches*. CA: Pyrczak Publishing

Transitory Provisions

10. As a living document, this treatise shall be updated and enhanced prescriptively. The university shall pursue an adoptive educational technology that reinforces academic standards and performance excellence tempered and guided by academic integrity and all the virtues therein for good conduct.

11. The various pedagogical frameworks shall continue to evolve as a matter of continuous quality improvements (CQIs), including the changing or evolving learning competencies and outcomes as demanded by regulatory and industry standards.

12. Applicable sanctions for breach of academic integrity and honesty provided for students, teachers, and non-teaching employees shall be imposed the same, and shall be reviewed for continuous improvement to adjust to emerging exigencies.

13. The Academic Affairs and the HRMD shall design continuous faculty and staff retooling in the application of AI in various disciplines and operational functions in the university, including the adoption of online programs.

14. The Academic Affairs, the RPC, and the HMRD shall continue to collaborate and promote tools and resources for research integrity, such as but not limited to *Literati Dos* (2023), that are being used by teachers and students.

15. The Academic Affairs in collaboration with the colleges and the LIC, shall explore and propose putting up of AI laboratory in the library for the use and exploitation of all students, faculty, and staff, in coordination with relevant disciplines.

16. This treatise shall be deployed and disseminated to all the stakeholders in the university as part of the official documents and shall be used for exhibit in all accreditation and certification evidences.

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SIGNED:

Dr. GUILLERMO P. TORRES, Jr.
University President

For distribution to all offices.